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Corrosion behavior of nickel-based superalloys in thermal storage medium of molten eutectic NaCl-$MgCl_2$ in atmosphere

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ABSTRACT

As a solar energy storage medium at moderate-to-high temperature, molten chlorate severally corrodes metal vessels. The corrosion behavior and mechanisms of pure nickel (Ni), GH4033 and GH4169 in 793 K molten eutectic NaCl-$MgCl_2$ were studied. During the initial stage of corrosion, pure Ni and GH4033 gain weight from the many plate-like shells formed on the sample surface. After corrosion for 40 h, all sample surfaces were rather porous and loosened. MgO nuclei that were attached to the loosened microstructure surface fell off with the microstructures, which led to a continual decrease in the sample weight. The three sample surfaces were mainly composed of MgO and Ni or (Ni, Fe) from the sample substrate. Al and Ti were preferentially oxidized and stably excited on the GH4033 surface, which inhibits the corrosion process. The corrosion rate of GH4169 was the highest for the δ phase, which is rich in Nb, and phase boundaries were corroded preferentially. In the anode reaction, metal atoms escape from the corrosion system in the form of chlorides or aqueous chlorides.

1. Introduction

As a phase change material (PCM) for solar energy storage media, molten chloride not only has the advantages of high chemical stability, low cost and high latent heat of solid-liquid phase change, but also the temperature of the solid-liquid phase change is high enough to satisfy the temperature needed for drying food in dusty environments and solar power [1,2]. In addition, molten chlorite is a candidate for use as a cooling media in nuclear reactors [3]. Compared with other mixtures of chlorite, eutectic NaCl-52 wt% $MgCl_2$ (melting point: 718 K, fusion heat: 430 J/g) is the best candidate material due to its higher latent heat value and lower vapor pressure [3,4]. A schematic of a pit solar energy collector-storage device, in which eutectic NaCl-$MgCl_2$ is the PCM, is shown in Fig. 1 [5]. The device proposed by the author of this paper has gained a patent. After being reflected by the first reflect mirrors and second reflect mirrors successively, focused solar radiation passes through the transparent manhole cover and enters the solar pit. A heat-transfer medium with high heat conductivity is heated and rises along several heat transfer tubes. PCM in the thermal storage tank is heated until fusion. Finally, the solar energy is stored. The temperature of the molten NaCl-$MgCl_2$ can be adjusted by adjusting the area of the first reflect mirrors. A heat exchange tube is embedded in the molten salt PCM. Steam flows in the heat exchange tube at low temperature and flows out at higher temperature to run the steam turbine and generator. As a PCM, eutectic NaCl-$MgCl_2$ can also be used in devices, which have been reported in the literature [6,7]. However, molten chlorite has a sharp corrosive action on metal thermal storage tanks and tubes [8–10].

Hot corrosion from NaCl and other chlorides is a hot research topic. Films of NaCl-Na_2SO_4 will deposit on critical surfaces of boiler tubes, gas turbines, and industrial waste incinerators, which will lead to hot corrosion on the alloy surface of Fe-based [11,12], Ni-based [13–17], Co-based [16] and ceramic coatings [17,18]. A basic flux model can largely explain the corrosion besides sulfuret mechanism. Oxyacid salt decomposes to Na_2O at high temperature. A protective oxide film will be dissolved by alkaline Na_2O, which is the basic flux model [16,19]. Further study shows that NaCl and other chlorites are the major cause of hot corrosion from NaCl-Na_2SO_4 [12,20]. The corrosion behavior of a single molten chlorite film of NaCl or KCl on a Fe-Cr alloy and pure Cr in air were studied by Y. S. Li, et al. [21–23]. The corrosion mechanism is explained by the oxide layer flux model or active oxidation mechanism. The primary reason is that the protective film of Cr_2O_3 is destroyed. In particular, the corrosion rate increases when certain corrosion products are formed, for example, $K(Na)_2CrO_4$ or its

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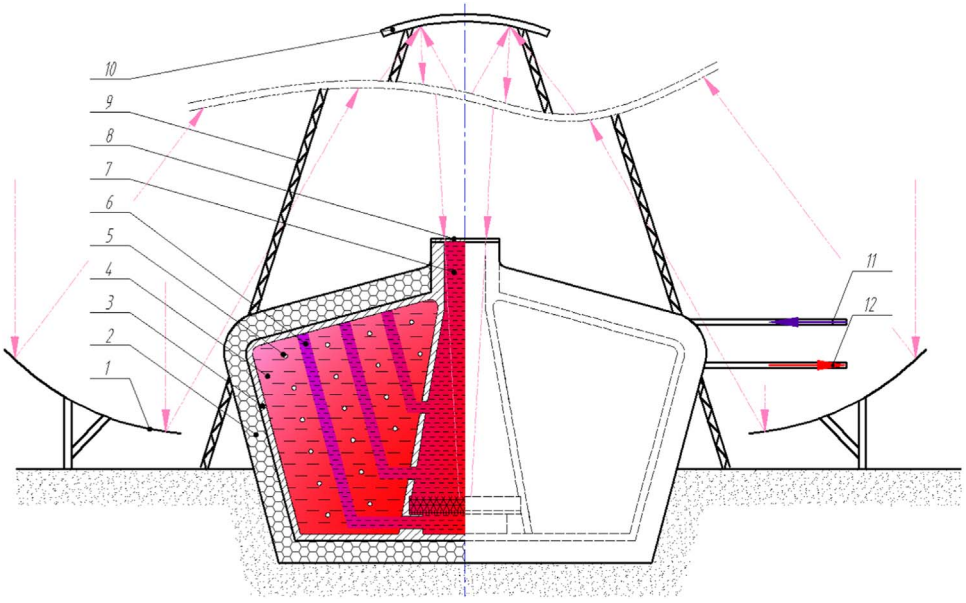


Fig. 1. Schematic of a pit solar energy collector-storage device. 1: First reflect mirrors, 2: Insulating layer, 3: Thermal storage tank, 4: PCM, 5: Heat exchange tube, 6: Heat-transfer medium, 7: Solar pit, 8: Manhole cover, 9: Support, 10: Second reflect mirrors, 11: Steam with low temperature, 12: Steam with high temperature [5].

mixture with alkali chloride, because corrosion products with low melting points are media for transporting oxygen from the air/salt interface to the salt/metal interface or for transporting corrosion products from the reaction region to the corrosion layer surface [21,23]. Alloy elements of Al and Si inhibit corrosion for its selective oxidation [21,24]. Either way, whether metal is corroded by a single molten chlorite film or a mixture of chlorite and oxyacid salt, the oxygen partial pressure is high. An oxidation film will be formed continuously, which result in corrosion behavior occurring continuously. Oxidation film flux is not the only cause of hot corrosion by alkali metal chlorides. In molten LiCl-KCl protected by an argon atmosphere, the corrosion rate of Yb is three times higher than that of Gd. In addition, there is a replacement reaction between Gd, Yb and K^+, Li^+ [25]. In molten LiF-NaF-KF (FLiNaK) protected by an argon atmosphere at 1123 K, intergranular corrosion occurs for Ni-based superalloys, for example, Haynes-230 and Inconel-617 [26]. Studies on the corrosion behavior of metal in molten eutectic NaCl-$MgCl_2$ are scarce. In addition, the corrosion mechanism is not clear in molten salt with low oxygen partial pressure.

Nickel-based superalloys are widely used under high-temperature working conditions for their advantageous processability and mechanical properties [13–15]. Therefore, taking GH4033 and GH4169 as research subjects, which have very different contents of Nb and Al, their corrosion behaviors in molten eutectic NaCl-$MgCl_2$ in atmosphere at 793 K were studied. Compared with pure Ni, the effects of alloy elements (Nb, Al, Ti and Cr) and second phases on the corrosion behavior were examined as well.

2. Experiment

Casting alloys of GH4033 and GH4169 were made under the protection of an argon atmosphere using a vacuum induction melting furnace (type: SP-25TC) based on electrolytically pure metals (purity > 99.9%; nickel (Ni), iron (Fe) and chromium (Cr)). The major compositions are shown in Table 1. Pure Ni, GH4033 and GH4169 are denoted 1#, 2# and 3#, respectively. Then, heat treatments were conducted as follows. GH4033 was heated to 1353 K (5 K/min), maintained at that temperature for 8 h, cooled to 973 K in air, and then maintained at that temperature for 16 h. Finally, it was cooled to room temperature in air. GH4169 was heated to 1223 K (5 K/min), maintained at that temperature for 40 min cooled to 993 K in air, and then maintained at that temperature for 8 h. Before cooling to room temperature, it was cooled to 893 K in a furnace, and maintained at that temperature for 8 h. All samples were cut into cubes with dimensions of 10 mm×10 mm×3 mm by an electrical spark linear incising machine tool under water cooling conditions. For other processes for preparing the samples, see reference [27].

Table 1
Compositions of samples [28] (wt%).

No.	Name	Fe	Ni	Cr	Mo	Al	Ti	Nb
1#	Ni	—	> 99.97	—	—	—	—	—
2#	GH4033	4.0	Bal.	20.5	—	0.8	2.6	—
3#	GH4169	18.5	Bal.	19.0	3.1	0.4	0.9	5.2

The eutectic salt mixture was made from anhydrous NaCl and $MgCl_2$. The NaCl purity was greater than 99.5 wt%. The registered trademark of the NaCl is “Shuangshuang”, which was provided by Shuangshuang, Ltd., of Yantai. The highest content of the main impurities in NaCl was as follows (wt%): $Fe(CN)_6$, 0.0001; K, 0.02; Ca, 0.05; Fe, 0.0002; Mg, 0.002; sulfate, 0.002; phosphate, 0.001; iodide, 0.002; and bromide, 0.01. The anhydrous $MgCl_2$ purity was greater than 99 wt%. The registered trademark is “Aladdin”. The $MgCl_2$ was provided by Sinopharm Chemical Reagent Beijing Co., Ltd. Before mixing the salt, NaCl was dried in a drying oven (DHG-9140A, Shanghai Jinghong Laboratorial Apparatus Co., Ltd.) at 393 K for 48 h. $MgCl_2$ was not dried, as it is highly hydroscopic. Then, 12.0000 g NaCl and 13.0000 g $MgCl_2$ were weighed by an electronic analytical balance with an accuracy of 0.1 mg (SOPTOP-FA2004, Shanghai Sunny Hengping Scientific Instrument Co., Ltd.), mixed extremely well and put in one corundum crucible in a glovebox with argon protection. Three mixture salts were prepared following the above process in three corundum crucibles. Because the salt mixtures are highly hydroscopic, the corundum crucibles were removed from the glovebox and quickly transferred to a heated muffle furnace (SX2-8–13, Xinhan Instruments Equipment Co., Ltd.). In addition, in order to lessen the absorption of moisture during the heating process, the muffle furnace was heated to 793 K from room temperature before the corundum crucibles were transferred. The heating rate was 5 K/min. After fusion of the eutectic salt, the four parallel samples, which were weighed (by an electronic analytical balance, SOPTOP-FA2004) and had their surface area

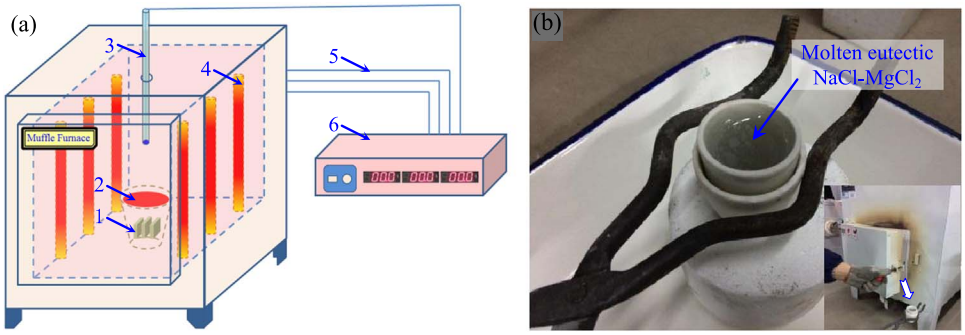


Fig. 2. Experimental setup (a) and molten eutectic NaCl-$MgCl_2$ removed from the setup (b). 1: Samples, 2: Molten salt, 3: Thermocouple, 4: Silicon carbide rod, 5: Wire, 6: Temperature controller.

measured (by a vernier caliper, Hengliang-530-150, accuracy 0.02 mm, Shanghai Hengliang Measuring Tool Co., Ltd.), were immersed in molten salt. Each kind of metal was put in one crucible. The experimental setup contained a box muffle furnace heated by silicon carbide rods. Photographs of the setup and molten eutectic NaCl-$MgCl_2$, which were removed from the setup after being melted at 793 K, are shown in Fig. 2(a) and (b), respectively. Because this dissertation aims to analyze the corrosion behavior of metals in molten salt in atmosphere, environments for the corrosion of molten salt under argon protection were avoided.

After corrosion for 20 h, the crucibles with the salt samples were removed from the furnace and cooled to room temperature. One sample of each kind of material was stored in a vacuum drying dish for later use. Other samples were cleaned by the mechanical method (following GB/T 16545-1996 [29]) as follows: all samples were ultrasonically cleaned in distilled water and ethanol for 10 min and 5 min, respectively, and then dried with cold air. After the samples were weighed by an electronic balance (SOPTOP-FA2004), they were placed in freshly melted salt, as previously mentioned. Then, the second corrosion experiment started. The above corrosion process was repeated eight times (160 h). However, the samples from the final corrosion experiment were not cleaned in water and ethanol. The method for cleaning the corrosion products followed the "ultrasonic cleaning method" in ASTM G1-03 [30]. The average weight change and associated errors were calculated following ASTM G1-03 and ASTM G16-95 [31], respectively. After corrosion testing and cooling to room temperature, profile photographs of the samples before and after ultrasonic cleaning as stated above were taken and are shown in Fig. 3(a) and (b), respectively. After cleaning, the 1# (pure Ni) surface is still relatively light, while 2# (GH4033) and 3# (GH4169) display a dim metallic color.

After corrosion for 160 h, the samples used to calculate the corrosion rate were first ultrasonically cleaned by the above-mentioned method and then soaked in hydrochloric acid solution (150 ml HCl in 1000 ml distilled water) for 2 min at 293 K [30]. Then, the samples were again ultrasonically cleaned in distilled water and ethanol for 10 min and 5 min, respectively. After drying with cold air, the samples were weighed by an electronic balance (SOPTOP-FA2004). The corrosion rate (V) was calculated by the following equation [30]:

$$V = \frac{K.\ W}{A.\ T.\ D} \tag{1}$$

where K is a constant [8.76×10^7 μm/y]; W is the mass loss of the sample during the corrosion test [grams]; A is the sample area [cm^2]; T is the exposure time of the corrosion test [hours]; and D is the sample density [g/cm^3]. The density of pure Ni, GH4033 and GH4169 is 8.9020 g/cm^3, 8.3924 g/cm^3 and 8.3507 g/cm^3, respectively.

To analyze the sample surface after corrosion in molten salt, one of the samples was prepared by the immersion method in water as follows. The sample was first quietly immersed in distilled water after cooling to room temperature. After thoroughly dissolving residual salt on the sample surface, the sample was carefully transferred to alcohol and then dried with cold air. During the process, the sample surface should be protected from damage. To observe the cross-sectional corrosion layer morphology, the second sample, which was corroded for 160 h, was prepared by the dry grinding method. The dry grinding method can avoid the loss of water-soluble corrosion products. For the third sample, which was corroded for 160 h, corrosion products on its surface were detected by a Rigaku D/max-2500/PC X-ray diffractometer (XRD) with a Cu Kα radiation source (λ=0.1541 nm). The tube voltage and current were 40 kV and 200 mA, respectively. Before XRD detection, residual salt on the sample surface was removed with a wood plate. The elemental distribution and cross-sectional and surface morphology were analyzed by a scanning electron microscope (SEM,

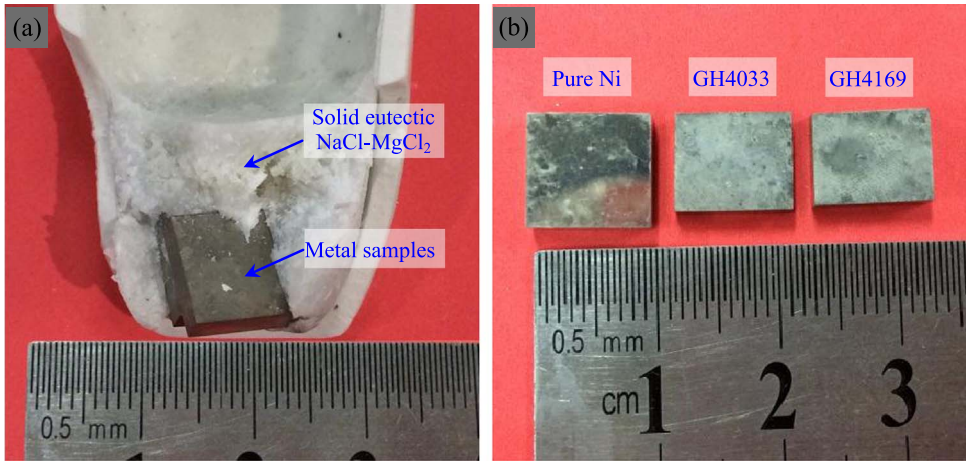


Fig. 3. After corrosion testing, profile photographs of the samples before (a) and after ultrasonic cleaning (b).

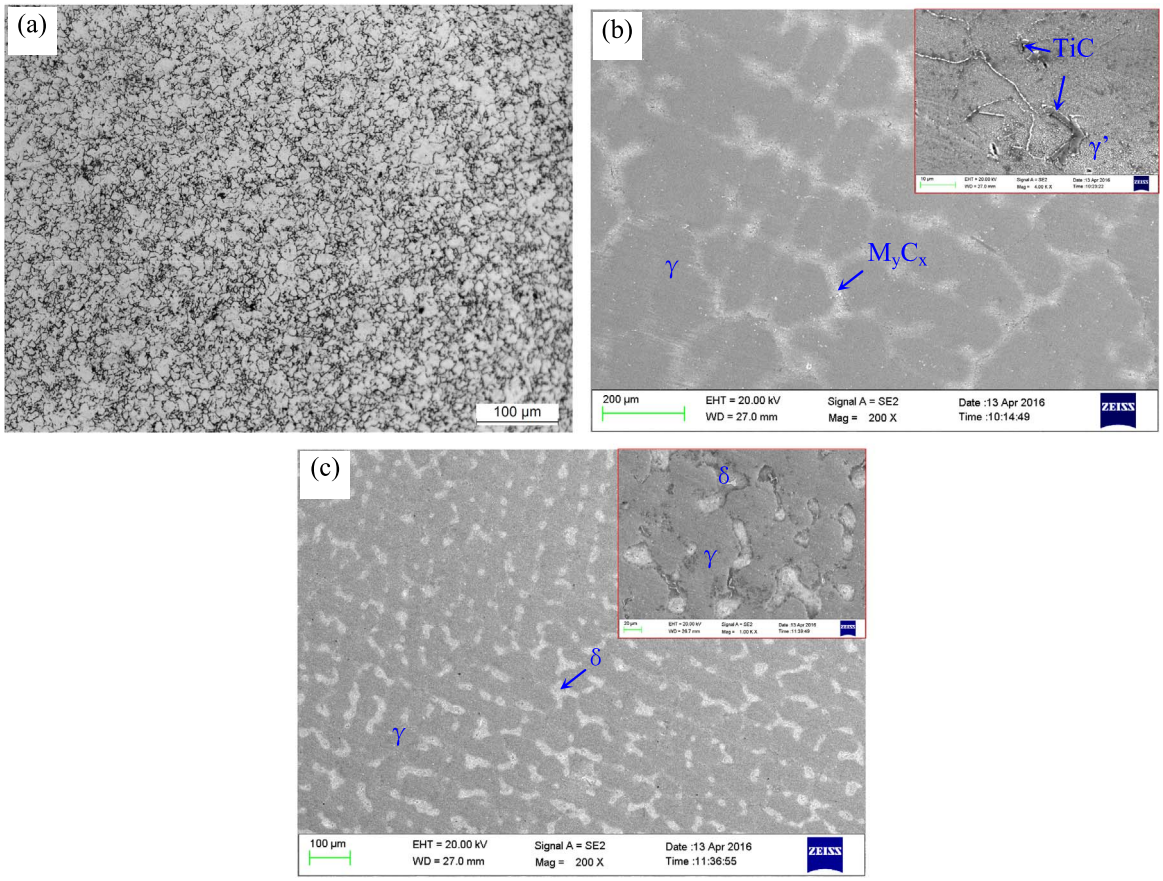


Fig. 4. Photographs of the microstructure and partial magnifications of the three samples: (a) pure Ni, (b) GH4033, and (c) GH4169.

ZEISS MERLIN Compact and JSM-6610 LV) with an attached energy-dispersive spectrometer (EDS).

3. Results and discussion

3.1. Experimental results

After heat treatment, for GH4033, there are dispersion γ' phase in substrate (γ) and carbide of M_yC_x (M = Ti, Cr, etc.) distributed along grain boundary. The primary phase of TiC and Ti(C, N) distributes along the grain. A bulk δ phase (the composition is nearly $(Ni_{0.96}Cr_{0.03}Fe_{0.01})_3$ $(Nb_{0.73}Ti_{0.12}Al_{0.10}Mo_{0.05})$), carbide and nitride will precipitate from the substrate (γ) [28]. Photographs of the microstructure and partial enlargement of pure Ni, GH4033 and GH4169, which were used in this experiment are shown in Fig. 4(a), (b) and (c), respectively. Grains of pure Ni are small and uniform. Based on the analysis of EDS, the carbide along the grain boundary in GH4033 is $(Cr, Ni, Fe)_{23}C_6$ and $(Cr, Ni, Fe)_7C_3$. The agglomerate is TiC. The γ' phase of $(Cr, Ni, Fe)_{23}C_6$ and $(Cr, Ni, Fe)_7C_3$ is distributed around the agglomerate. For GH4169, the many large δ-phase sheets are mainly composed of Ni and Nb precipitates at the grain boundary.

The corrosion rate, at a 95% confidence interval, of pure Ni, GH4033 and GH4169 is 57.0 ± 9.0 μm/y, 141.9 ± 11.2 μm/y and 246.4 ± 13.4 μm/y, which are shown in Fig. 5(a). The relationship between the average weight change and exposure time is shown in Fig. 5(b). Overall, using the same cleaning conditions for all samples, the mass of all the samples always decreases. However, at the initial stage, the mass of pure Ni and GH4033 decrease first and then increase slightly. At 40 h, their masses reach a maximum and then decrease continuously. The reasons for the phenomenon are discussed later in this paper.

After corrosion for 20 h and preparation by the immersion method in water, microstructure pictures of pure Ni, GH4033 and GH4169 are shown in Fig. 6(a), (b) and (c), respectively. There are many chink along the grain boundary on the pure Ni sample surface and a porous structure on the GH4033 and GH4169 surfaces. In addition, an incomplete shell is formed on the surface of all the samples. Based on the EDS analysis of the shell and XRD analysis of the sample surface, which will be shown below, the major composition of the shell is MgO with a small amount of Ni.

The porous structure on sample substrate has little MgO. In addition, the shell on the pure Ni surface is bigger and thicker than on other samples. Granular carbide was clearly seen on the GH4033 surface as well. Fig. 6(d) shows the EDS results of the shell and substrate on the GH4169 surface.

The surfaces of pure Ni, GH4033 and GH4169 are greatly loosened after corrosion for 160 h and preparation by the immersion method. Morphology photographs of the surfaces are shown in Fig. 7(a), (b) and (c). The grain boundary on pure Ni is deep. In addition, many grains are exposed. The surfaces of the other samples are granular shaped. A large shell, like the shell that formed after corrosion for 20 h, was not be found on any of the sample surfaces after corrosion for 160 h.

The XRD results of pure the Ni, GH4033 and GH4169 surfaces from Fig. 6 and Fig. 7 are shown in Fig. 8(a), (b) and (c). For all of the samples, the major phase is Ni and/or (Ni, Fe), which is the sample substrate, and a certain amount of MgO is also seen. Trace amounts of oxides (M_3O_4) of Fe, Cr and Ni were also detected for GH4033 and GH4169.

Water-soluble or alcohol-soluble corrosion products may be lost during the cleaning process in the immersion method. The phase on the sample surface (after 160 h corrosion) was reexamined by XRD after the residual salt on the surface was peeled off by a wood plank. The results are shown in Fig. 9. As shown in Fig. 8, the major phases on pure Ni and GH4033 are Ni and MgO and on GH4169 are (Ni, Fe), γ'

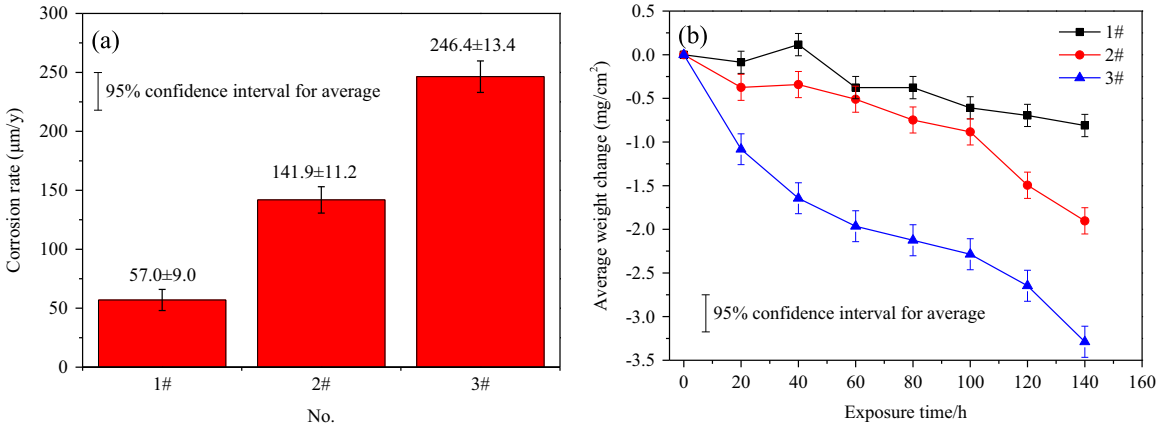


Fig. 5. Corrosion rate (a) and average weight change versus exposure time (b).

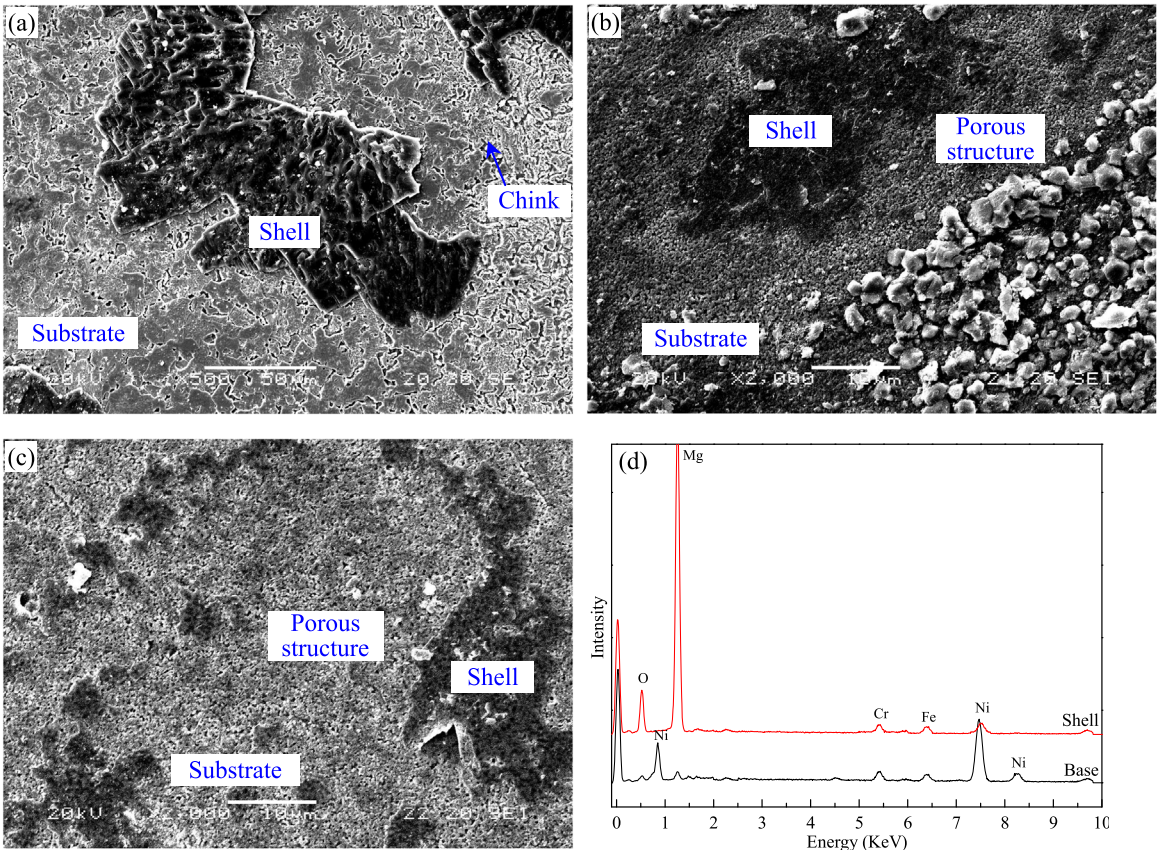


Fig. 6. SEM photos and EDS results of the three samples after corrosion for 20 h and prepared by the immersion method in water: (a) pure Ni, (b) GH4033, (c) GH4169, and (d) the EDS result.

(FeNi$_3$) and MgO. In addition, other compounds were found, including Na_2MgCl_4, $NaMgCl_3$, $MgCl_2{\cdot}6H_2O$, and NaCl. Na_2MgCl_4 and $NaMgCl_3$ are the eutectic reaction products of $MgCl_2$ and NaCl. $MgCl_2{\cdot}6H_2O$ is the product of $MgCl_2$ absorbing moisture during the salt melting process. Certainly, H_2O may also be introduced into the salt mixture by being adsorbed on the sample surface. In the phase diagram of NaCl-$MgCl_2$, the point of the acuter composition moves left after $MgCl_2$ absorbs moisture. So, NaCl is pro-eutectic products during salt solidification. Trace amounts of metal oxides were found on the surface of GH4033 and GH4169. Studies examining only metals in molten eutectic NaCl-$MgCl_2$, which avoid effects of atmosphere (moisture, oxygen, etc.), should been done next.

After corrosion for 160 h, cross-sectional photographs of pure Ni, GH4033 and GH4169 are shown in Fig. 10(a), (b) and (c). It is found from Fig. 10(a) that the surface is unevenness, and a corrosion layer is not clearly observed. For GH4033, there is a corrosion layer, which is approximately 21 μm in depth at most sites and 40 μm at the localized position. The corrosion layer is divided into outer region I and inner region II in Fig. 10(b). Region I is compact, and region II is porous. There is no other phenomenon associated with the small amount of dispersed carbide (M_yC_x). For GH4169, a corrosion layer with a maximum depth of approximately 75 μm, shown in Fig. 10(b), is formed near the surface. In addition, the δ phase was preferentially corroded. Many large corrosion holds were connected and formed a crack along the phase boundary. The corrosion layer is thin at the localized location and is approximately 33 μm. However, the corrosion holes are larger.

Fig. 11(a) and (b) gives the EDS results of the sample cross section

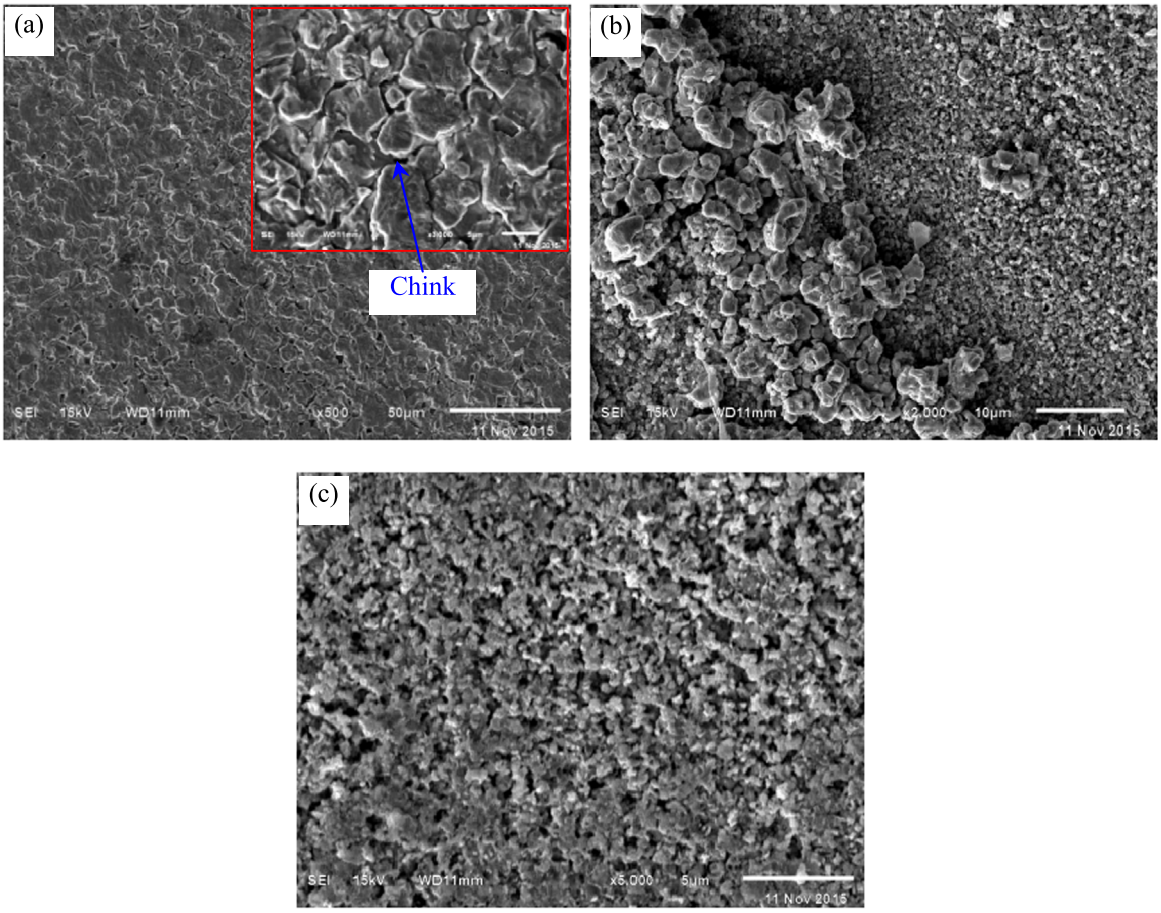


Fig. 7. SEM photographs of the three samples after corrosion for 160 h and prepared by the immersion method: (a) pure Ni, (b) GH4033, (c) GH4169.

near the corrosion layers shown in Fig. 10(b) and (c). Compared with the content of Cr in the substrate, it is found from Fig. 11(a) that the content of Cr is decreased in region II and increased at the bottom of region I. The content of Ni showed a contrasting trend to that of Cr, but the content of Ti increased sharply only at the bottom of region I. The change in elements Al, Mg and O is similar to that in Ti. Therefore, oxides of Cr, Al, Mg and Ti may be formed in region I. Changes in Fe are not clear, probably because of the low content in the GH4033 substrate. In Fig. 11(b), the content of Cr and Fe decreased and that of Ni increased near the corrosion layer. The content of other elements, including Nb, Mo, Ti, Al, Mg, Cl and O, vary slightly with less character.

There are similar compositions of residual salt that corrode the three samples after 20 h. The XRD results are shown in Fig. 12. The major compositions are NaCl, Na_2MgCl_4 and $NaMgCl_3$, along with a small amount of MgO and $MgCl_{22}O$). The reason for their formation is same as that of the salt crust on the sample surface, as previously mentioned. Oxides or chlorides of Ni, Cr, Fe, Ti and Nb were not found.

3.2. Discussion

Because molten eutectic NaCl-$MgCl_2$ is an electrolyte, Ni atoms on the grain boundary with disordered arrangement and high energy will be preferentially dissolved as anode. Then, Ni^{2+} forms [21]. Electrons lost from Ni atoms may be captured by O atoms or Mg^{2+}. Then, Mg forms. The reactions are expressed as:

$$Ni \rightarrow Ni^{2+} + 2e \quad (2)$$

$$O + 2e \rightarrow O^{2-} \quad (3)$$

$$Mg^{2+} + 2e \rightarrow Mg \quad (4)$$

Consequently, as shown in Fig. 6(a) and Fig. 7(a), the grain boundary was preferentially corroded on the pure Ni surface. The mass of the sample decreased. Compared with Ni, the Gibbs free energy at which Mg is oxidized is more negative ($\Delta G^{o}_{NiO,793K}$ =−333.375 kJ/mol, $\Delta G^{o}_{MgO,793K}$ =−1032.4 kJ/mol). Therefore, Mg will be oxidized to MgO preferentially. NiO can be reduced to Ni by Mg even if Ni has been oxidized. Meanwhile, MgO may form when Mg^{2+} reacts with O^{2-} in molten salt. The above reactions are given by:

$$Mg + O \rightarrow MgO\ \Delta G^{\circ}_{793K} = -717.2kJ/mol \quad (5)$$

$$Mg + NiO \rightarrow MgO + Ni\ \Delta G^{\circ}_{793K} = -349.5kJ/mol \quad (6)$$

$$Mg^{2+} + O^{2-} \rightarrow MgO\ \Delta G^{\circ}_{793K} = -1184.6kJ/mol \quad (7)$$

MgO nuclei form in corrosion holes along the grain boundary and grow upwards on the smooth sample surface, as shown in Fig. 6(a). A schematic of the above process is shown in Fig. 13(a).

With the MgO nuclei growing upwards and thickening, a large MgO shell forms. However, MgO has low plasticity. In addition, the thermal expansion coefficients of MgO (13.8×10^{-6} /K) and pure Ni (17.6×10^{-6} /K) are very different (27.5%). The MgO shell may craze and peel off during cold and hot cycling. In addition, as corrosion holes penetrate the substrate along the grain boundary, the holes will finally connect. On the sample surface, grains and MgO nuclei that are attached will peel off from the substrate. The process of nucleation and peeling of MgO repeats on the sample surface. A schematic of the above process is shown in Fig. 13(b). The sample weight will increase when the weight of MgO gained is higher than the mass loss, which is caused by the solution of Ni atoms peeling off of grains and MgO. Therefore, the sample weight after corrosion for 40 h increased, which is shown in Fig. 5(b).

The sample surface will become rough as more grains fall off. On a

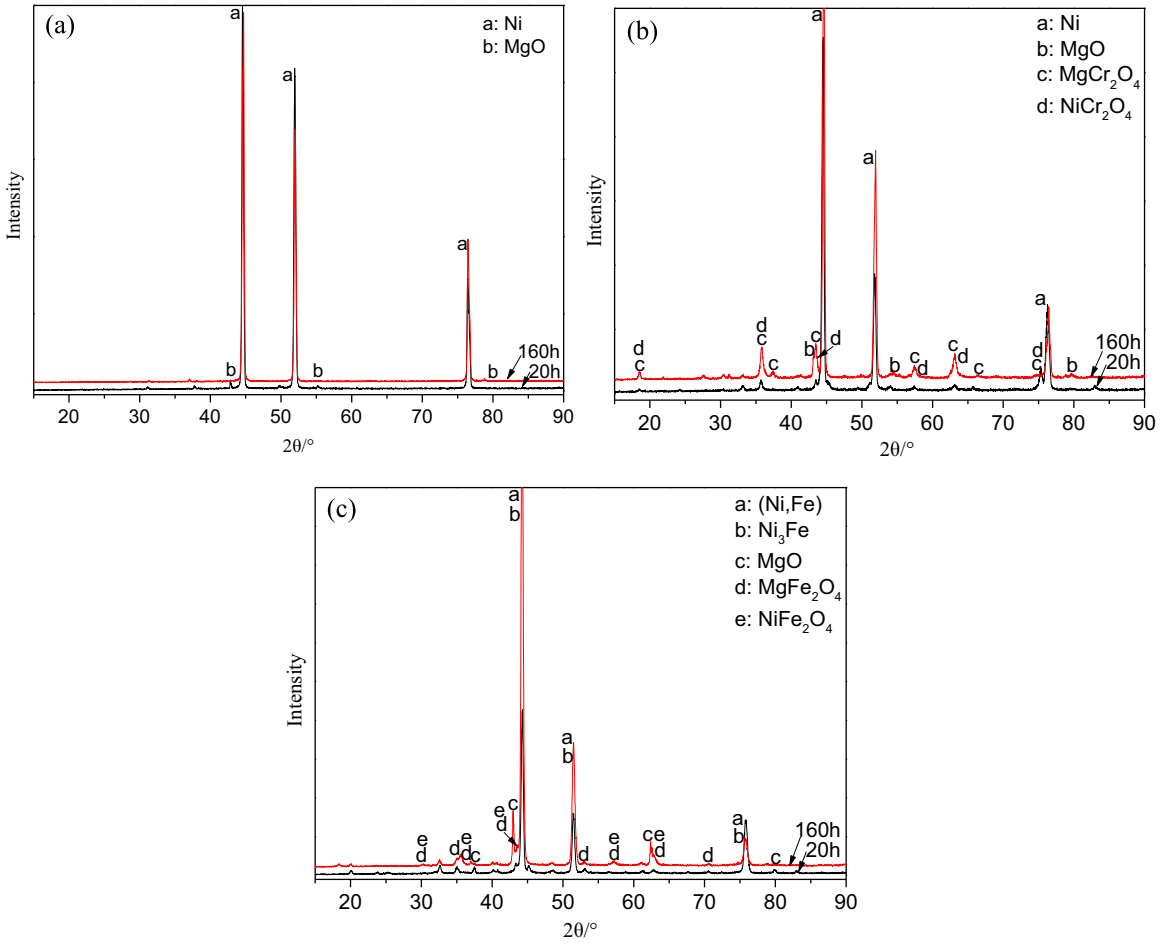


Fig. 8. XRD results of the sample surfaces after corrosion for different times and prepared by the immersion method in water: (a) pure Ni, (b) GH4033, (c) GH4169.

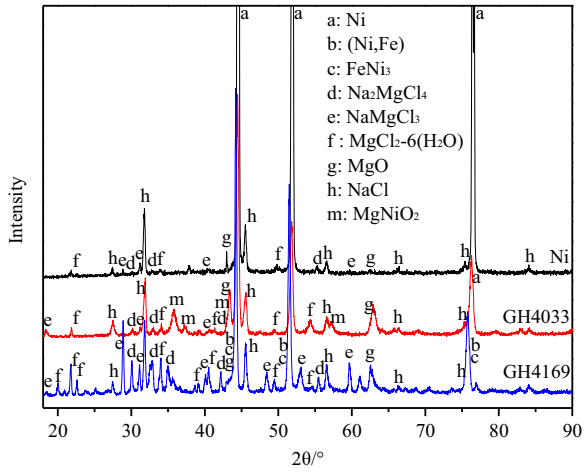


Fig. 9. XRD results of sample surfaces after corrosion for 160 h and peeling off of residual salt.

rough surface, it is difficult for MgO nuclei to grow into a large connected shell. MgO nuclei on a surface with isolated grains easily fall off. The process is shown in Fig. 13(c). Consequently, the mass of the pure Ni sample decreases continually after 40 h of corrosion. After 160 h, as shown in Fig. 7(a), the MgO shell is not found on the pure Ni surface. In Fig. 10(a), the sample surface is uneven.

GH4033 and GH4169 contain Fe at approximately 4 wt% and 18 wt %, respectively, and Cr at approximately 20 wt% in both. The electrode potential is different among the elements Cr, Fe and Ni. Elements with low electrode potential will be dissolved as shown in reaction (2). As shown in Fig. 11, on the corrosion layer, the content of Cr and Fe decreased due to dissolving, and many corrosion holes were formed. Molten salt penetrated into the corrosion holes. In addition, corrosion reactions (2–4) occurred continually. MgO formed on the sample surface as shown in reactions (5–7). At the initial stage of corrosion, as with the pure Ni sample as mentioned above, because the sample surface is flat, the MgO nuclei grow upwards and connect, which is shown in Fig. 6(b) and (c). The schematics of this process on GH4033 and GH4169 are shown in Fig. 13(d) and (g), respectively. The alloy samples gain weight from MgO deposition as well. Therefore, it is found from Fig. 5(b) that the mass of GH4033 after corrosion for 40 h increases slightly more than it does after corrosion for 20 h.

As corrosion continues, the microstructure of the sample surface shows a porous framework. The strength of the framework structure is low. MgO will fall off when the strength is so low that it cannot support MgO. After long-term corrosion, as with the pure Ni samples, the process of nucleation and peeling off of MgO repeats. It is difficult for MgO to grow upwards and connect. The above scheme of the process of GH4033 and GH4169 is shown in Fig. 13(e), (f) and Fig. (h), (i). Therefore, as shown in Fig. 8(b) and (c), for GH4033 and GH4169, after corrosion for 160 h, MgO (small nuclei) detected from the sample surface after preparation by the immersion method. However, the MgO shell is not found in Fig. 7(b) and (c). The delicate framework not only has low strength but also has a high corrosion rate for a larger surface area. So, as shown in Fig. 5(a) and (b), the mass of GH4033 and GH4169 decrease continually after corrosion for 40 h. Their corrosion rates are higher than that of the pure Ni sample by approximately 149% and 332%, respectively.

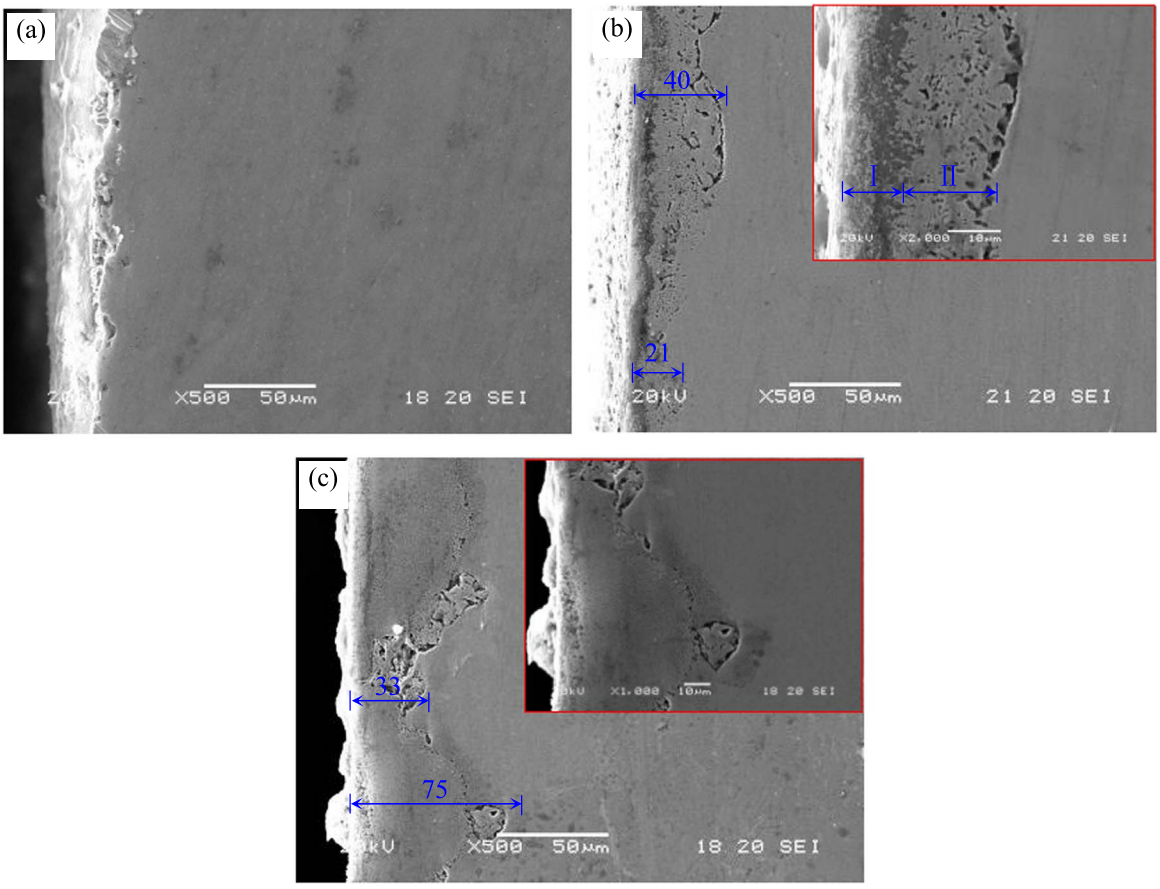


Fig. 10. Cross-sectional SEM photographs of the samples after corrosion for 160 h and dry grinding: (a) pure Ni, (b) GH4033, (c) GH4169.

In addition, because the surface area of GH4033 and GH4169 are larger after corrosion for 160 h, the combined strength of the salt crust, corrosion products and sample surface is high after solidification. Consequently, in the XRD results shown in Fig. 9, after corrosion for 160 h and clearing residual salt from the sample surface with the same method, the content of MgO, $MgCl_{22}O$) and other compounds on the GH4033 and GH4169 surfaces is higher than that on the pure Ni surface.

The Gibbs free energies of the oxidation reaction between the alloy elements in GH4033 and 1.0 mol O_2 at 793 K are shown in Table 2.

It is found from Table 2 that Cr and Fe will be oxidized preferentially, because the Gibbs free energy of their oxidized products is more negative than that of Ni [32,33]. The alloy elements Cr and Fe will diffuse from the sample substrate and the bottom of the corrosion layer (region II in Fig. 10(b)) to the corrosion layer surface (region I in Fig. 10(b)). Diffusion and preferential oxidation are the second reason that the content of Cr and Fe decrease and small corrosion holes form on the corrosion layer, as shown in Fig. 11(a). Table 2 shows that the Gibbs free energy when Al and Ti are oxidized is more negative than that of Fe and Ni. And the energy close to it of Mg is oxidized to MgO. According to reaction (6) and the correlation analysis above, Al_2O_3 and TiO_2 rather steadily exist as preferentially oxidation products. Therefore, the material may be Al_2O_3 and TiO_2 at the bottom of region I in Fig. 11(a). Region I has a high content of Al, Ti and O and a compact structure. A certain amount of Cr_2O_3 was mixed in the compact structure as well. The compact corrosion layer steadily exists when the grow rate is higher than the dissolution rate of the oxide layer. In addition, the corrosion process is inhibited in the compact

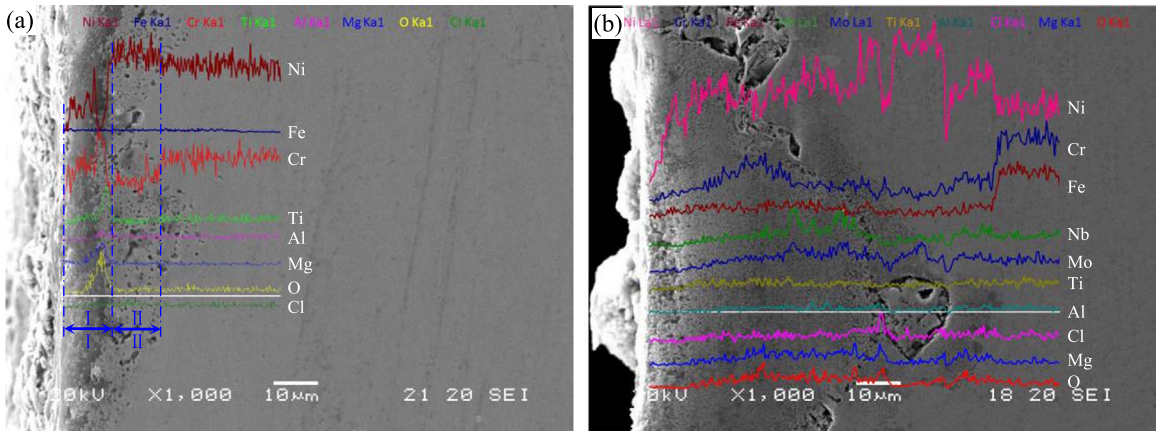


Fig. 11. EDS results of sample cross sections after corrosion for 160 h and dry grinding: (a) GH4033 and (b) GH4169.

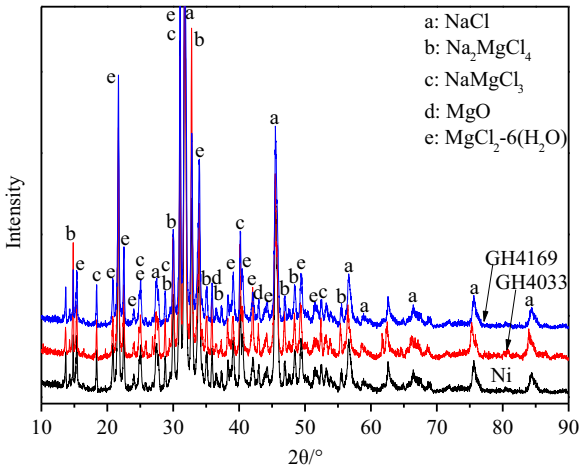


Fig. 12. XRD results of the residual salt after corrosion of the three samples for 20 h.

layer, slowing the mass transfer of molten salt. Schematics of the above process are shown in Fig. 13(e) and (f). In addition to other reactions with Cr_2O_3 in the molten salt system, because the solubility of Cr_2O_3 is high [34], the Cr_2O_3 layer was destroyed. In addition, the elemental content of Cr decreases on surface region I. It is speculated, based on the results of Fig. 11(a), that the Al_2O_3 and TiO_2 layers will be destroyed, just like Cr_2O_3, although reports on the solubility of Al_2O_3 and TiO_2 in molten chlorite could not be found. Therefore, the content of Ti and Al decrease on region I. In addition, the compounds Cr, Ti and Al were not found on the GH4033 surface after corrosion for 160 h.

There is little Ti and Al in GH4169, but it has many massive δ phases, composed of Nb and Ni, on the substrate (γ phase). It is found from Fig. 10(c) that not only the γ phase was corroded (a corrosion layer of approximately 33 μm formed) but also the δ phase was corroded preferentially. In addition, like the grain boundary, the phase boundary of δ and γ was preferentially corroded as well. Molten salt will penetrate into the sample substrate through corrosion holes, which promote the corrosion of the δ phase. As corrosion continues, the bonding strength between the δ phase and substrate γ phase decreases. When the strength is low enough, the δ phase with parts of the MgO nuclei/shell will flake off from the sample substrate. Consequently, the mass of GH4169 did not increase as other two samples after corrosion for 40 h but decreased quickly in all the corrosion processes. As shown in Fig. 5(a), the corrosion rate of GH4169 is the fastest in all samples. A schematic of the GH4169 corrosion mechanism is shown in Fig. 13(g), (h) and (i).

$MgCl_2$ will absorb moisture during the melting of the mixed salt in atmosphere. Then, $MgCl_{22}O$) is formed. $MgCl_{22}O$) will decompose into HCl at high temperature. HCl reacts with dissolved oxygen to generate Cl_2 and H_2O. Cl_2 will react with alloy elements, and volatile metal chlorides (MCl_n; M = Cr, Fe, Ni, etc.) will be formed [13,35]. The Gibbs free energies of generating $NiCl_2$, $CrCl_2$, $CrCl_3$, $FeCl_2$ and $FeCl_3$ are −185.4 kJ/mol, −296.006 kJ/mol, −259.168 kJ/mol, −241.111 kJ/mol and −166.029 kJ/mol, respectively. Thus, Cr and Fe will be chlorinated preferentially compared with Ni, which is the third reason for the decrease in Cr and Fe content and the formation of small corrosion holes on the corrosion layer. Aqueous chlorides with low melting points also formed after the chloride absorbed moisture.

$$4HCl(g) + O_2(g) \rightarrow 2Cl_2(g) + 2H_2O(g)\Delta G^{\circ}_{793K} = -10.1kJ/mol \quad (8)$$

$$nCl_2 + 2M \rightarrow 2MCl_n \quad (9)$$

$$MCl_n + H_2O \rightarrow MCl_n{\cdot}m(H_2O) \quad (10)$$

Metal particles, which peel off from the sample substrate, as shown in Fig. 13, or are formed through reaction (6), have large surface areas. The rate of reactions (9)-(10) is fast. Consequently, although the samples were obviously corroded, as shown in Fig. 6, Fig. 7 and Fig. 10, large amounts of oxides or chlorides of alloy elements Ni, Cr and Fe were not detected, except for traces of $MgCr_2O_4$, (Ni, Fe)Fe_2O_4 and $MgNiO_2$, as shown in in Fig. 8 and Fig. 9. Compounds of the alloy elements were also not clearly observed in the residual salt, which is shown in Fig. 12.

4. Conclusion

(1) Pure Ni, GH4033 and GH4169 are corroded severely in 793 K molten eutectic NaCl-$MgCl_2$, and their corrosion rates are 57.0 ± 9.0 μm/y, 141.9 ± 11.2 μm/y and 246.4 ± 13.4 μm/y, respectively. After corrosion for 20 h, pure Ni is corroded along the grain boundary. For GH4033 and GH4169, the surfaces are porous, and the corrosion layer is loose. The compounds on the three sample surfaces are mainly Ni or (Ni, Fe), which are from the alloy substrate, and MgO. Large amounts of oxides or chlorides of alloy elements Ni, Cr and Fe are not found. After corrosion for 160 h, a uniform corrosion layer of approximately 21 μm forms on the

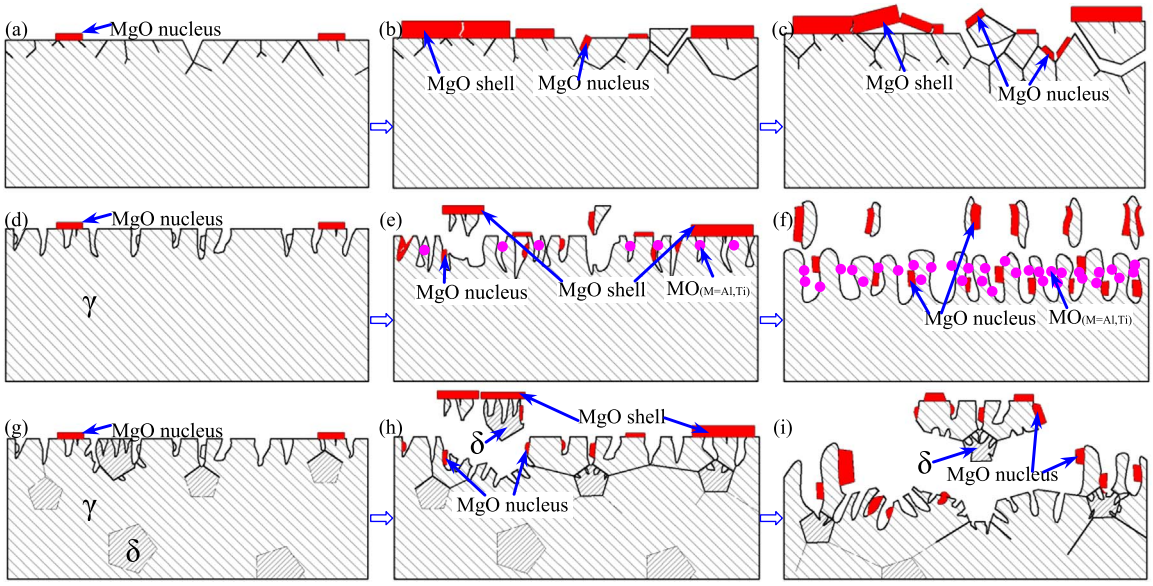


Fig. 13. Schematic for the corrosion mechanism of the three samples at different stages: (a), (b), (c) pure Ni, (d), (e), (f) GH4033, and (g), (h), (i) GH4169.

Table 2
Gibbs free energies of chemical reaction between alloy elements and 1 mol O_2 at 793 K (kJ/mol).

Oxide	NiO	Cr_2O_3	FeO	Fe_2O_3	Fe_3O_4	Al_2O_3	TiO	TiO_2	NbO	MgO
$\Delta G°$/(kJ/mol)	−333.4	−615.8	−425.0	−407.3	−425.6	−951.2	−931.1	−799.4	−694.1	−1032.4

GH4033 surface. For GH4169, the corrosion layer is nonuniform.

(2) During the initial stage of corrosion, a MgO shell forms on the smooth sample surface. The samples gain weight when the mass increase from MgO deposition is higher than the loss of weight from corrosion or MgO peeling off. After corrosion for 40 h, the sample surface shows a porous framework. MgO nuclei have difficulty growing into a shell and peel off with grains or loose corrosion layers, which leads to continual sample weight loss. The δ phase and phase boundary in GH4169 were corroded preferentially, and the sample mass decreased continually. Metal atoms dissolved from the three samples escaped from the corrosion system in the form of chlorides or aqueous chlorides.

(3) The corrosion mechanism is from the combined effects of the alloy dissolved as anode, preferential oxidization and chlorination. These are also the reasons why the content of Cr and Fe decrease and small corrosion holes form on the alloy corrosion layer.

(4) Because the Gibbs free energies of the oxidation reactions of Ti and Al are close to that of Mg. A compact and steady oxide layer formed on the corrosion layer of GH4033, with a high content of Ti and Al. The corrosion process was inhibited. However, for GH4169, the δ phase, which is rich in Nb, and phase boundary are corroded preferentially. The phase boundary is the mass transfer channel of molten salt, which accelerates corrosion.

(5) As a metal vessel for eutectic NaCl-$MgCl_2$ PCM, the corrosion rate of GH4033 and GH4169 are too high to meet the requirement of corrosion resistance. Although the corrosion resistance of pure Ni is relatively good in the experiment, the mechanical properties of the pure metal are less than satisfactory. Consequently, a material with excellent corrosion resistance in molten eutectic NaCl-$MgCl_2$ should be searched for continuously, and an advanced single-phase Ni-based alloy without Cr should be developed.

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